Faster Quantum Calibration with AI



Open_qblock

NCHC × OpenACC × NVIDIA Hackathon

4 November 2026

OpenACC More Science, Less Programming

OPEN HACKATHONS

NCHC 國家實驗研究院 國家高速網路與計算中心 National Center for High-performance Computing

NVIDIA

Open_qblock Team

Team members



錢祖晟
NTHU
Computer Science



彭國忠
NTU
Physics



李俊華
NTU
Physics



李建林
NTHU
Computer Science



徐義竣
NTU
Physics



胡凱翔
NTU
Electronics

Mentor



王勻遠
NVIDIA

Advisers

管希聖
NTU
Physics

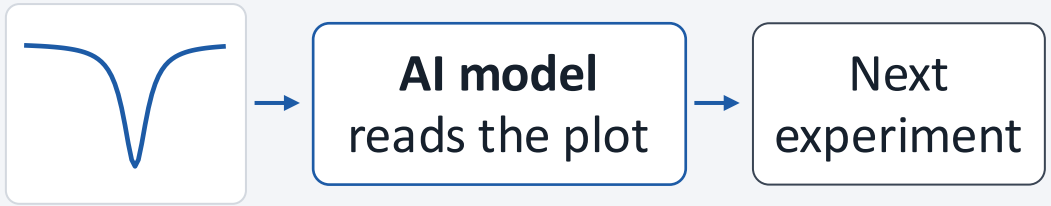
江介宏
NTU
EE

陳南佑
NCHC

Two Approaches to Automated Calibration

❶ Single-qubit setup

錢祖晟 · 李建林 · 胡凱翔

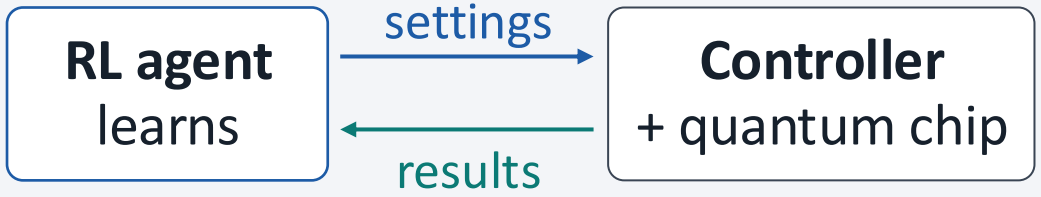


Calibration plot

- AI interprets calibration plots
- Qblox hardware and CUDA-Q Dynamics on H200
- NVIDIA Ising Calibration

❷ Entangling-gate calibration

彭國忠 · 李俊華 · 徐義竣



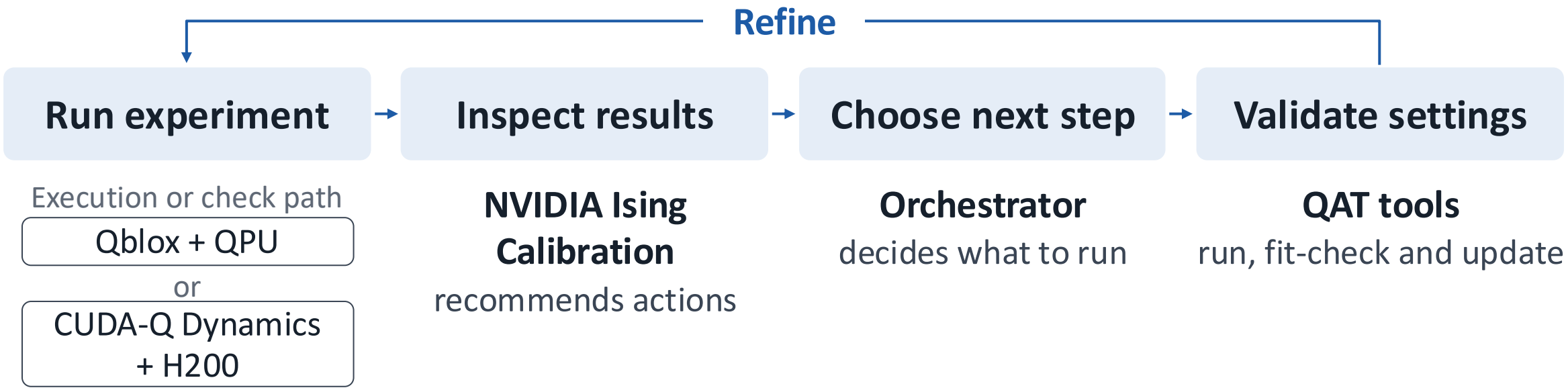
- RL learns control settings from measurements
- OPX1000 controller and NVQLink with OPNIC + GH200
- Bell-pair and four-qubit state tests

Shared goal: shorter calibration turnaround

Automating Single-Qubit Setup

Workflow architecture

Prototype · no benchmark results



12-experiment reference workflow



Some routines measure properties (for example T1) rather than write new settings.

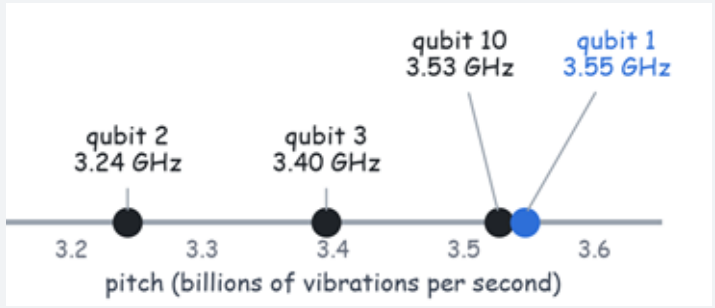
Quantum calibration: tuning every qubit before it can compute

1 A qubit rings like a tuning fork



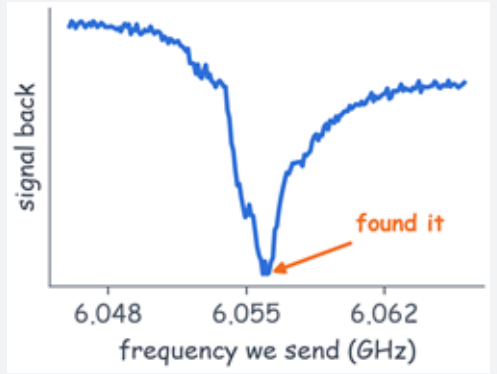
- A tiny circuit, colder than outer space
- Has its own pitch, like a tuning fork
- A short microwave hum at that pitch flips 0 to 1

2 No two qubits are alike



- Same design, different pitch
- Like tuning every string of a new guitar
- So each qubit is measured on its own

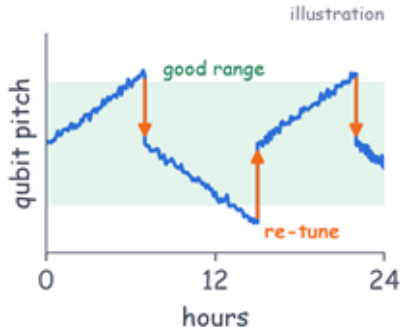
3 Calibration finds the settings



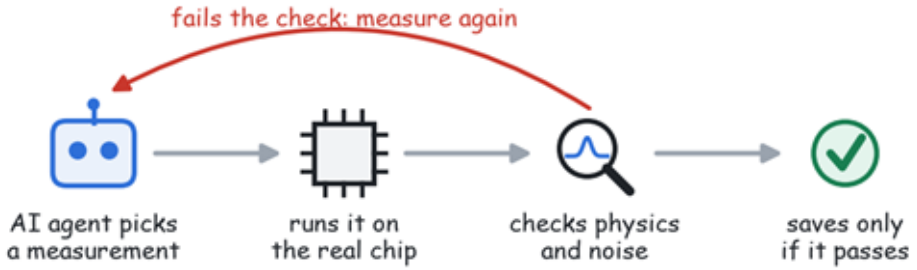
- Sweep the frequency, watch what comes back
- Like turning a radio dial to find a station
- The dip is the setting we keep

Calibration matters

- A small error spoils every result
- **No alarm goes off**
- Settings drift: re-tune often
- By hand: ~1–1.5 h per qubit pair*



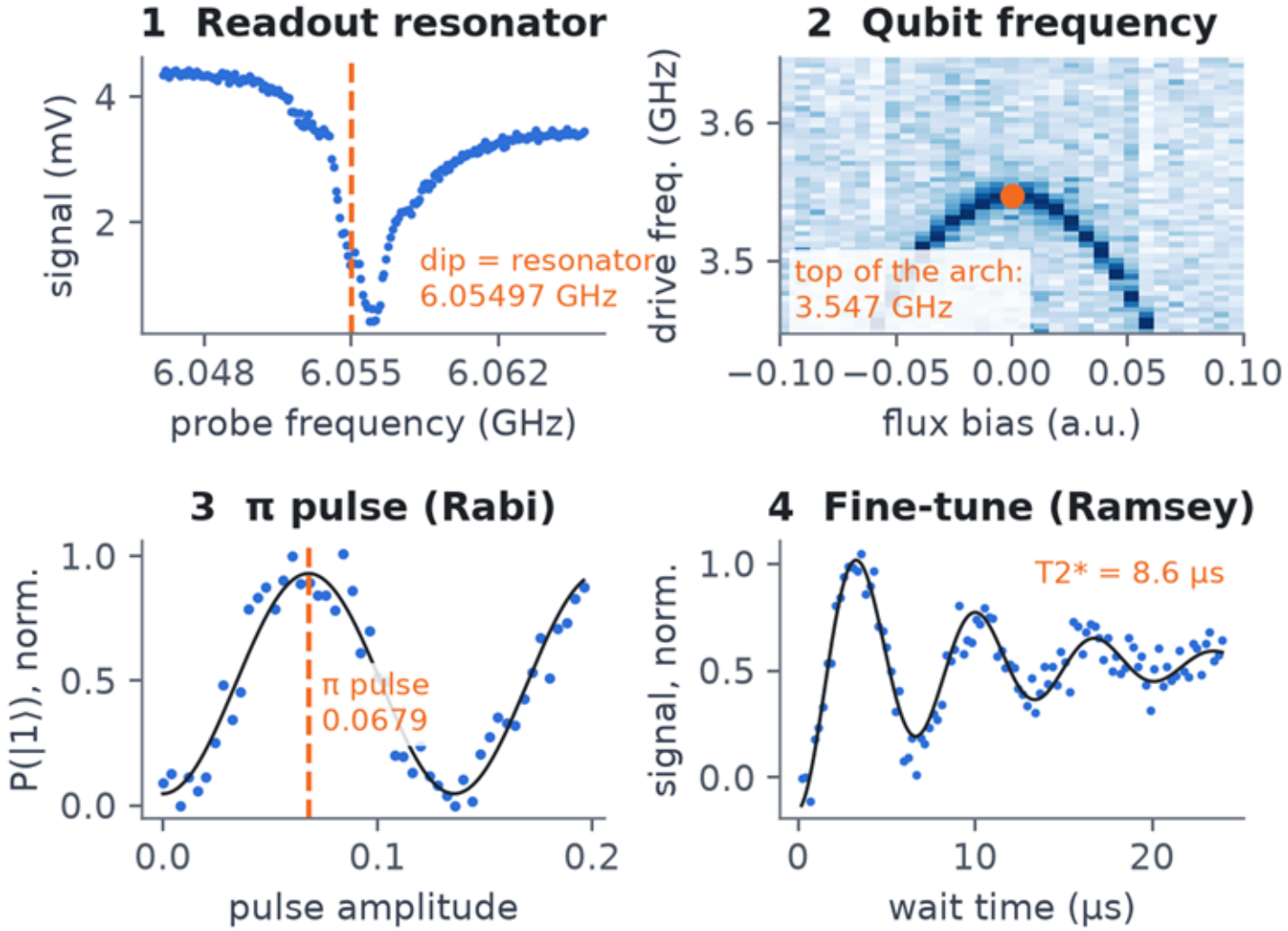
The workflow our team built



*Estimate. Krantz et al., Appl. Phys. Rev. 6, 021318 (2019). Data: qubit q1 on our 10-qubit chip, 21 Oct 2026.

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How we measure on one real qubit



Sweep configurations, fit the model, then keep the number

1 **Readout:** 6.055 GHz
2 **Qubit frequency:** 3.547 GHz
3 **Pulse strength:** 0.068
4 **Fine-tune:** 0.35 MHz correction

- Data must fit the physics shape
- Signal at least 5× the noise
- Every number has an error bar
- Different experiments must agree

Krantz et al., Appl. Phys. Rev. 6, 021318 (2019); Kelly et al., arXiv:1803.03226 (2018). Data: qubit q1, 10-qubit chip, 21 Oct 2026.

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Simulation first on CUDA-Q

All steps pass

15 of 15

- 19 minutes, no API cost
- No human step needed

Validity Rate

95.5%

- 95.5% of final numbers within tolerance
- 0 wrong numbers saved

Ising Model checks plots

93%

- NVIDIA Ising Calibration reviews each plot
- Agrees with the physics check 93% of the time

Simulate first

- No chip time or booking needed
- We know the true answer, so we can grade the agent
- Bugs found here never reach the real chip

Same loop as chip simulating

Simulated qubit (CUDA-Q) → AI agent picks a measurement → Saves only if it passes

Simulator: QAT CUDA-Q backend, simulated qubit q1. Tolerance: qubit frequency ±0.5 MHz, readout ±1 MHz, π pulse ±2%.

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Almost all models can run the whole simulation

Agent	Plot reader	Steps passed	Time	API cost
Opus 5.5	Opus 5.5	14 of 15	9 min	$0.88
Opus 5.5	NVIDIA Ising ★	14 of 15	9 min	$0.53
Nemotron VL ★	Nemotron VL ★	15 of 15	19 min	$0
Beam-1.0-501B	—	13 of 15	6m 24s	—
Nemotron Ultra 550B	—	13 of 15	9m 36s	—

- Diverse LLMs can successfully execute nearly the entire calibration simulation workflow
- Demonstrating the feasibility of autonomous calibration even with smaller, more cost-efficient models.

★ runs on NCHC GPUs. Normal-signal simulated qubit.

Simulator: QAT CUDA-Q backend, simulated qubit q1; one run per row, Opus rows before the punchout fix. Data: 3–4 Nov 2026.

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Real QPU calibration: time to a valid qubit frequency

Main agent / plot reader	Calibration time	Ising inspecting time	Run calls	Non-bypass nodes
Opus 5.5 / Ising	28:38	11:09	12	4 Valid, 5 tries
Muse Spark Open / Ising †	19:15	3:28	10	1 Valid 3 Failed, 6 tries
Muse + skill / Ising	43:54	14:18	27	4 Valid, 18 tries
Human (Me) + Asking GPT	About 45 mins	—	19	4 Valid, 12 tries

- **Separate reasoning and vision contexts still supported successful calibration:**

  Ising interpreted the plots, while Opus combined its feedback with numerical results and playbooks.
- With additional workflow guidance and checkpoints, Muse + Ising also passed all four essential nodes.

† Failed run ends at its stop, so its shorter time does not indicate faster calibration. Single runs with differing conditions.

Source: QAT index + OpenCode JSONL. Time in min:sec.

Real QPU calibration: token usage and recorded cost

Main agent / plot reader	Cache read + write (M)	Output (k)	Total (M)	Total Cost (USD)	Estimated main-agent cost (USD)
Opus / Ising	4.857	25.559	4.882	$2.964	$1.48
Muse + skill / Ising	5.496	40.985	5.537	—	$1.06

- **Ising model intergration overhead still needs optimization**

  Processing Ising responses involves substantial cached-context reads, increasing overall token usage.

- **Guidance from stronger model could help the weaker perform better**

  Muse completed all four required calibration nodes

  Estimated main-agent cost was ~29% lower with Muse + Ising than with Opus + Ising (1.4× cost reduction).

All input and cache tokens were merged and priced at cache-read rates to normalize provider reporting differences. Output and reasoning tokens were priced at output rates. Estimates use public API prices, exclude Ising costs, and do not represent actual RAP charges.

Source: per-turn OpenCode token and cost records. Values rounded. Diagnostic launches and later SSH failures excluded.

Muse + Ising: calibration succeeds with workflow support

Strategy / result	Earlier run (OpenRouter)	Latest run (NCHC RAP)
Recovery guidance	Some guidance base on QAT playbook	Explicit diagnosis and complete recovery inputs (scan range, points, amplitudes, repetitions)
Tool enforcement (Ising plot inspecting)	Prompt reminders only	Runtime checkpoint requires explicit inputs and a parseable Ising review
Signal recovery	5 invalid qubit flux spectroscopy attempts Stopped before Rabi	Spectroscopy: drive/readout 0.1, 1000 reps Ramsey: repetitions increased from 500 to 2000
Non-Bypass nodes	**1 / 4 passed** **6 attempts, 1 valid**	**4 / 4 passed** **18 attempts, 4 valid**

Opus-derived guidance and runtime checks supported all four required calibrations

All 4 non-bypass nodes valid within the first 8 steps

Several conditions changed together, including provider and human-assisted SSH recovery / restart.
The earlier run already had partial Opus guidance.

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Key Findings

40% less time than human + GPT assistance

Opus + Ising: 28.6 min. Human + GPT reference: ~45 min.
And cheaper then me…

Ising supported successful delegation of plot interpretation

Opus + Ising passed all 4 essential nodes, using 12 experiment calls vs. 14.
Opus combined Ising's visual feedback with numerical results and playbooks.

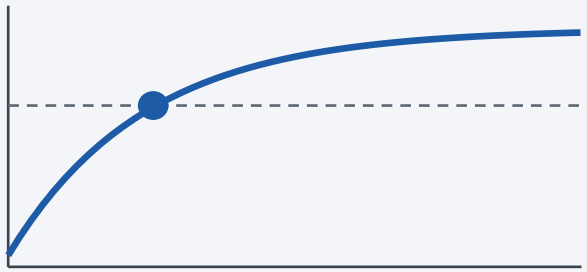
Workflow guidance helped models complete essential calibration

Opus-derived recovery guidance and runtime checkpoints supported
4/4 essential nodes passing, versus 1/4 in the earlier failed run with Muse model.

Single-run comparisons through Step 8 under differing conditions.
RAP pricing and Ising token costs remain unmeasured. All 4 essential nodes passed does not mean all 8 steps passed.

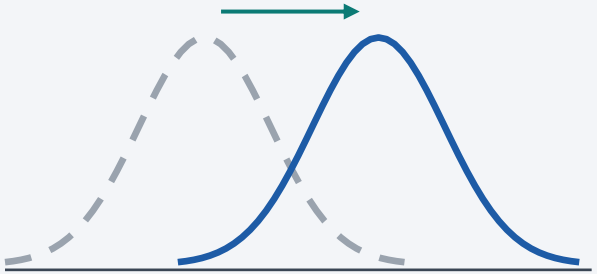
Gate Calibration: Workflow and Results

Reinforcement learning and NVQLink with OPNIC + GH200



1 Calibration speed

How much turnaround time do we save?



2 Drift during operation

What changes while the device runs?



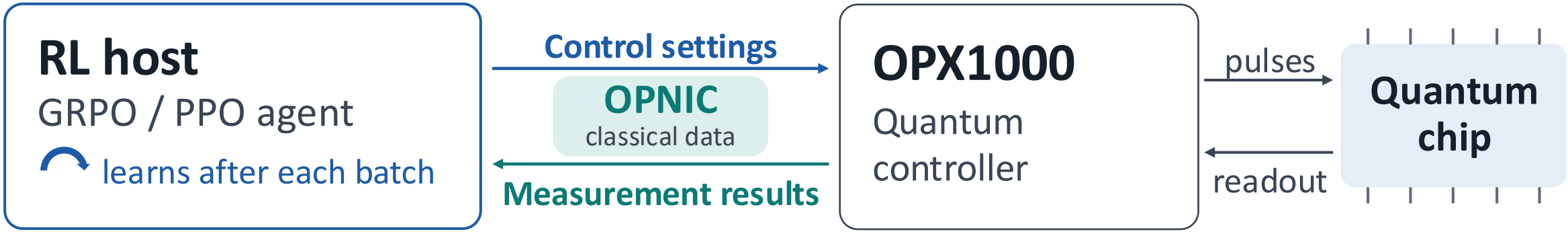
3 Four-qubit performance

What carries into a larger circuit?

Illustrations are conceptual, not measured data.

A Faster Feedback Loop for Gate Calibration

Platform: **NVQLink with OPNIC + GH200**



≈100–120×

Faster application communication

Deployment

Remote API via tunnel vs colocated OPNIC stack

Learning and scoring ran on CPU

One batch = 20 candidate settings

Remote API: **≈492–584 ms** → OPNIC: **≈4.9 ms** per batch

Prior Work Supports Learning-Based Calibration

PRIOR WORK · PEER-REVIEWED

Li et al., Physical Review X (2024)

99.90 ± 0.01%

Leakage-aware average CZ gate fidelity

48 ns gate

Peer-reviewed hardware experiment

PRIOR WORK · VENDOR-REPORTED

Academia Sinica / Quantum Machines (2026)

≈15 min to 25 s

Two-qubit gate calibration, manual vs RL

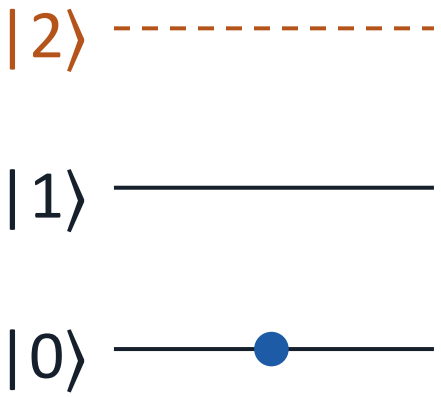
≈36× from the vendor-reported times

Vendor-reported comparison; TD3 agent

Our tests evaluate this approach on our device.

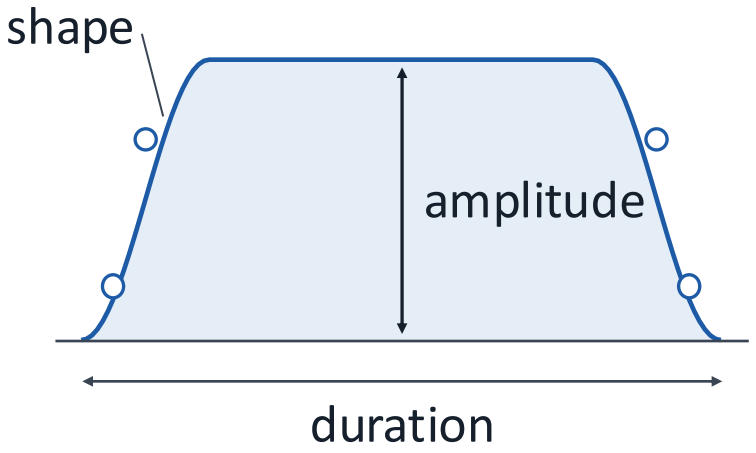
Control Pulses Shape the Quantum Operation

❶ Prepare



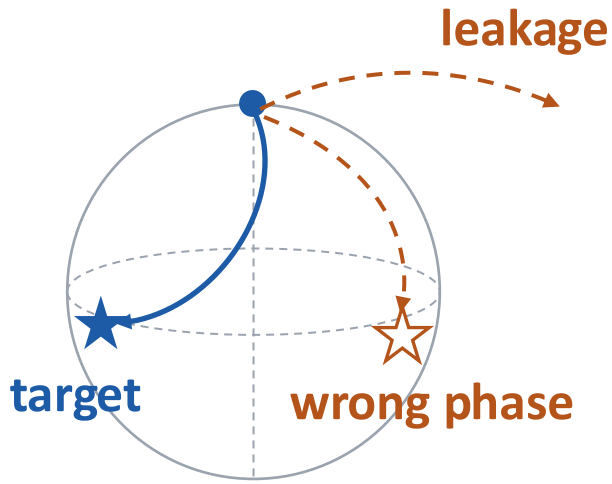
|0⟩, |1⟩: qubit levels
|2⟩: unwanted level

❷ Apply a shaped pulse



Pulse shape controls the operation

❸ Reach the target



Conceptual single-qubit picture

Leakage and phase errors reduce performance

Two Ways We Shortened Calibration

1 Recorded workflow

25.9×

Faster recorded workflow

Manual **19.4 min** → agent **45 s**

2 Communication

≈100–120×

Faster application communication

Remote API→
NVQLink with OPNIC + GH200

Different baselines and timing scopes: the two results are never multiplied.

Two Qubits, One Bell State

One shared quantum state



$$|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$$

Prepared with local rotations and a CZ pulse

Each outcome is random.
One result predicts its partner in the chosen measurement view.

Measure both along the same axis

Results: **0/1** along Z, **+/−** along X or Y

	Ideal Bell pair Φ^+	Coherence erased (00/11 mixture)
Z	Same **0 0** or **1 1**	Same **0 0** or **1 1**
X	Same **+ +** or **− −**	Uncorrelated
Y	Opposite **+ −** or **− +**	Uncorrelated

Matching 0/1 outcomes alone does not prove entanglement.

Other measurement bases reveal the shared coherence.

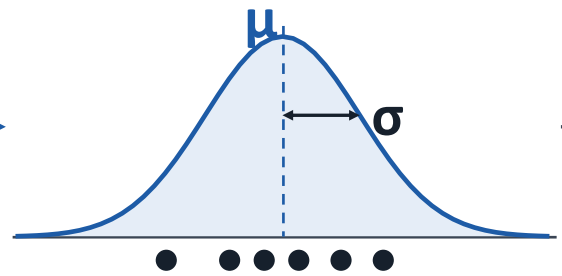
Why Bell pairs matter

Gate tests · Teleportation · Larger entangled states

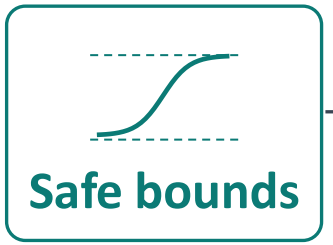
The Agent Learns Four Gate Controls

One mean and standard deviation per control

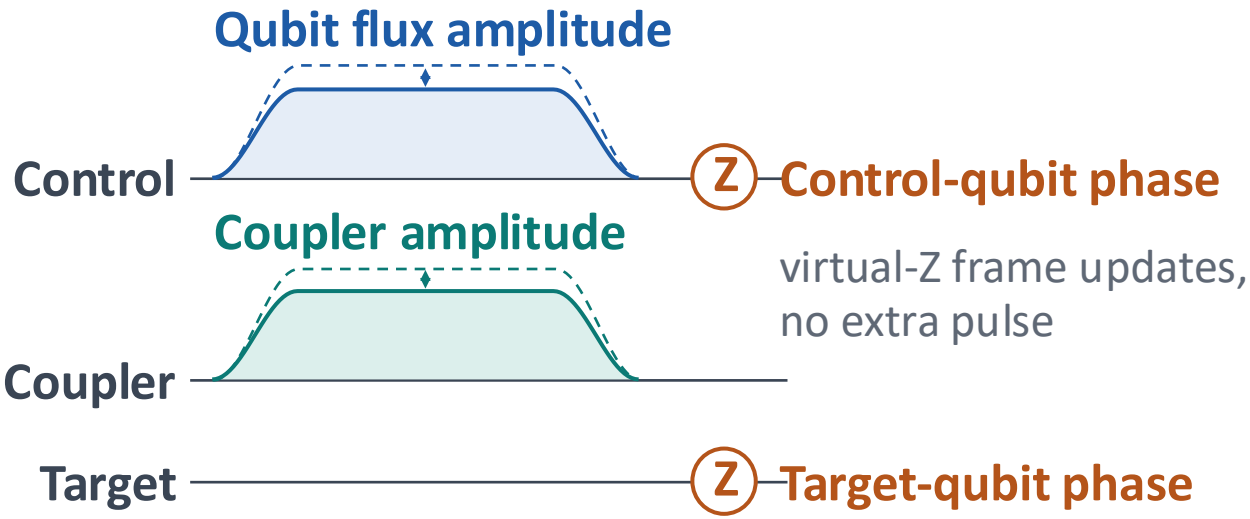
❶ Sample settings



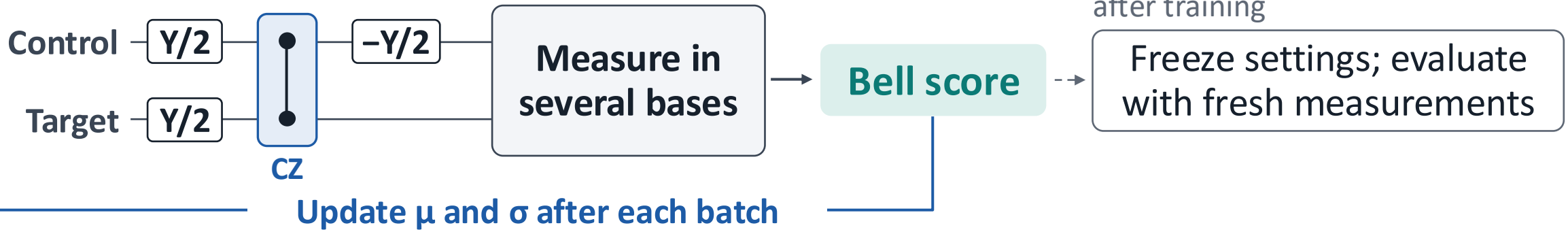
μ: mean
σ: exploration width



❷ Apply them to the stored gate

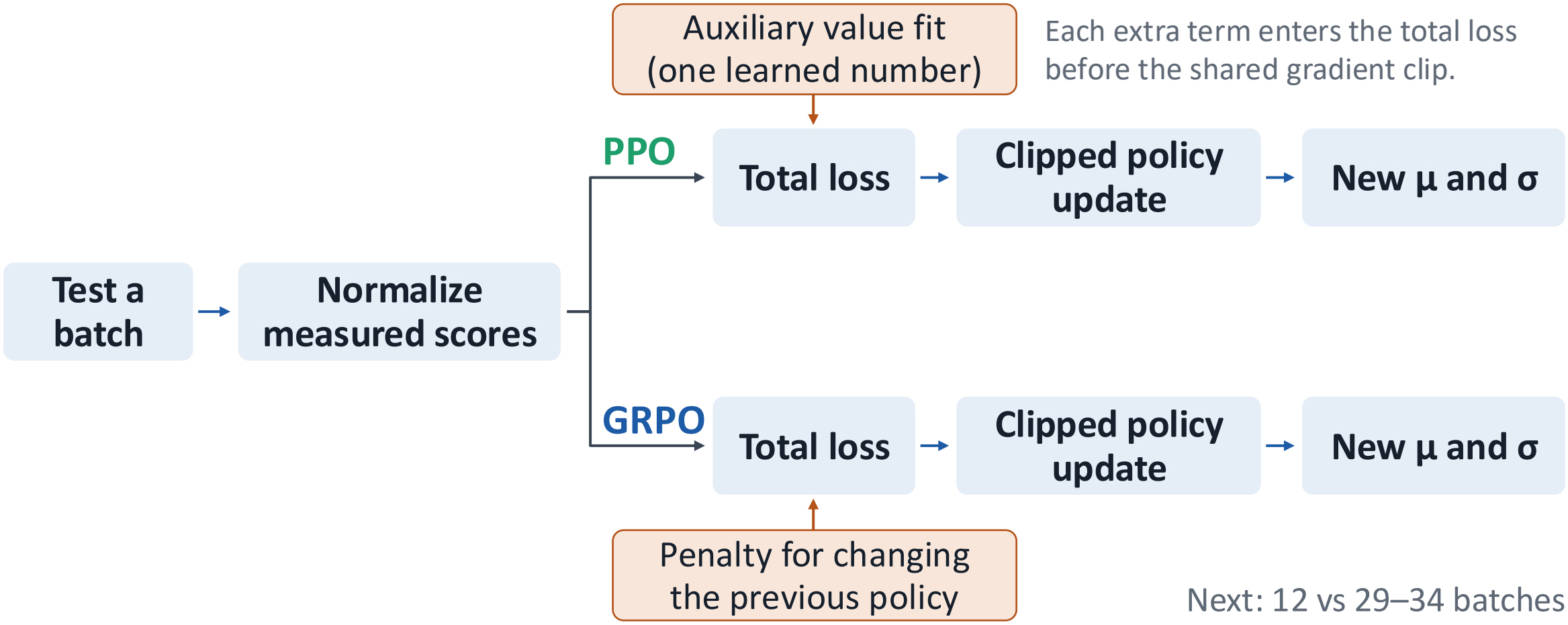


❸ Score with a Bell test



How PPO and GRPO Update the Agent

Our one-step implementations

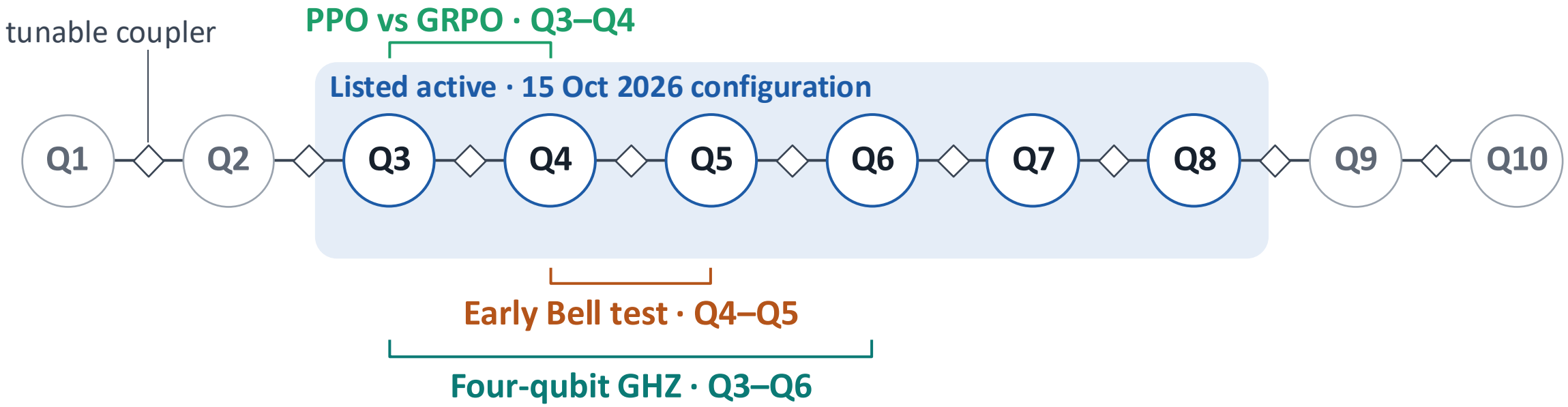


= **Same score normalization**

≠ **Different extra terms in the update**

The Processor and the Tested Qubits

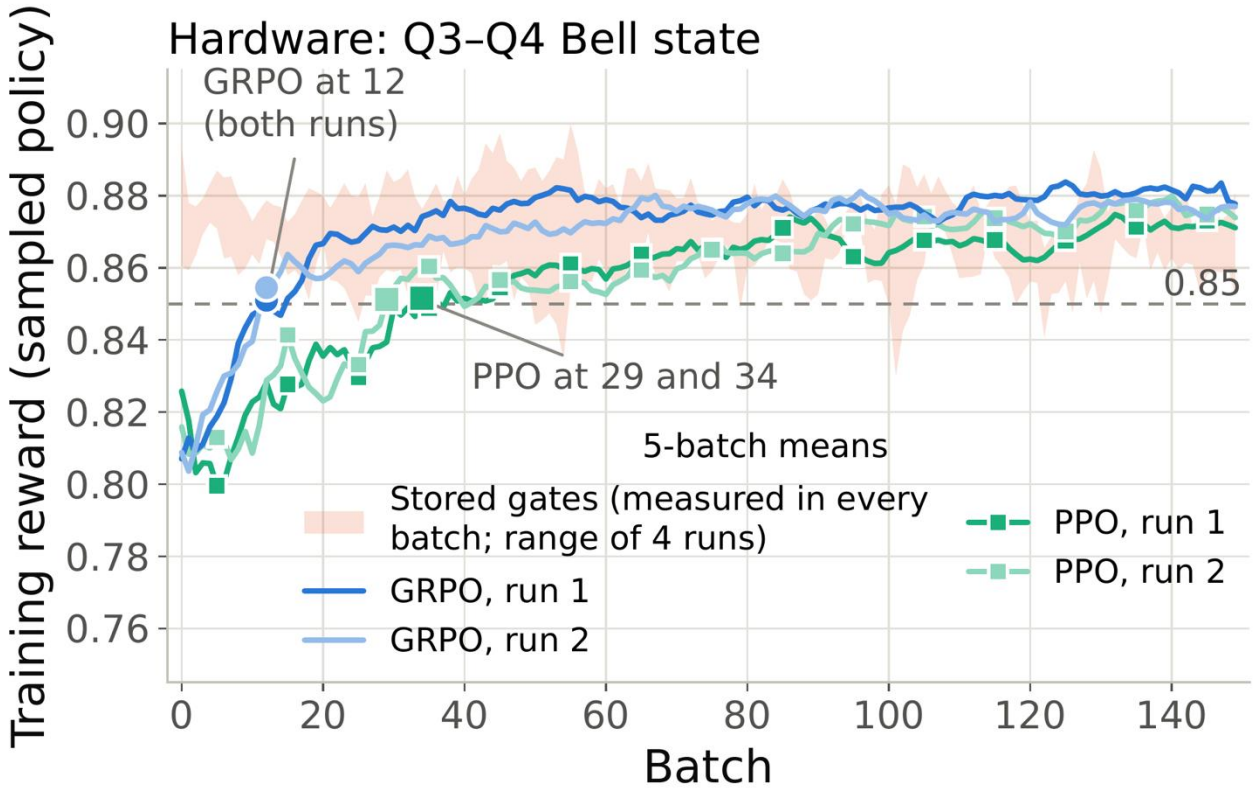
Connectivity schematic



- 10 superconducting qubits
- Tunable couplers between neighbours
- OPX1000 controller

Configuration snapshot: 15 Oct 2026 · Tested paths shown; connectivity is schematic

GRPO Reached the Training Threshold in Fewer Batches



Earlier reward improvement;
similar held-out Bell-state fidelity.

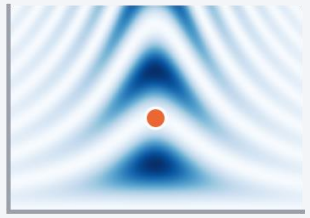
2.4–2.8×

PPO / GRPO batches to the training threshold

GRPO: 12 batches **PPO:** 29 and 34

Hardware learning-rate schedules differed.

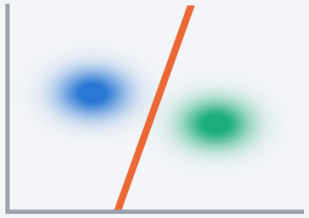
Manual Calibration Includes Experiments and Decisions



Tune amplitude



Adjust phases



Update readout



Verify Bell state

Thumbnails are schematic

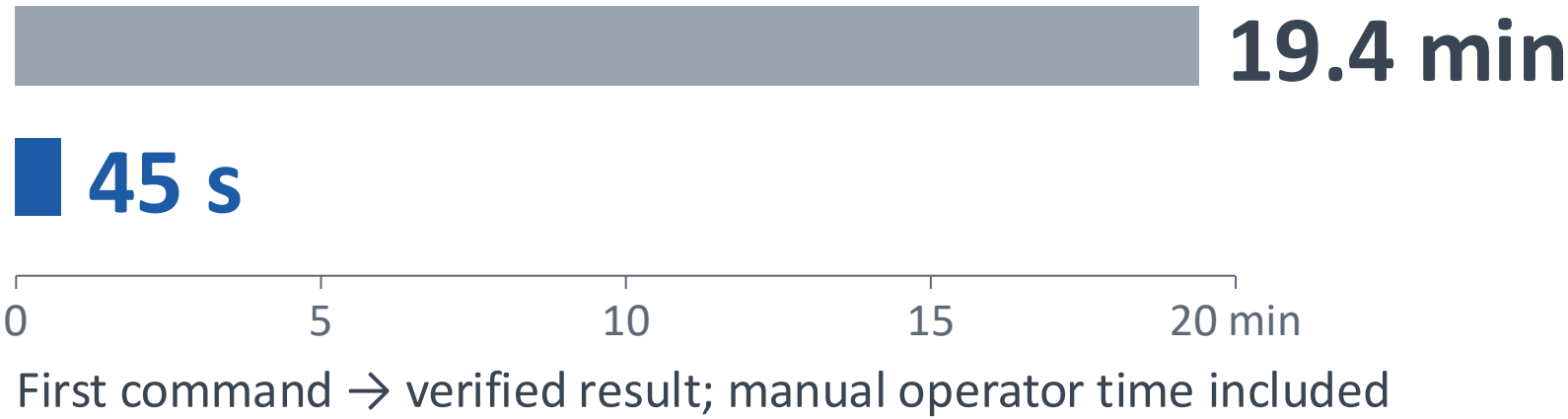
Measurements alternate with reading fits and entering commands

19.4 min

Recorded manual workflow, including operator time

From 19.4 Minutes to 45 Seconds

25.9× **faster recorded workflow**



First command → verified result; manual operator time included

15–30-shot workloads: 37–45 s including final verification

q5–q6, 15 shots: **37.1 s** | q3–q4, 30 shots: **45.0 s**

Means of complete agent runs per workload, not a same-pair shot sweep

Recorded: manual 20 Oct, one session · agent 3 Nov, mean of six runs

GRPO Learning Across Three Qubit Pairs

Remote API **OPNIC:** NVQLink with OPNIC + GH200 ◆◆ diamond = 2000-shot held-out final

solid = mean, band = ±1 SD -- -- dashed = best candidate - - - gray dashed = manual reference

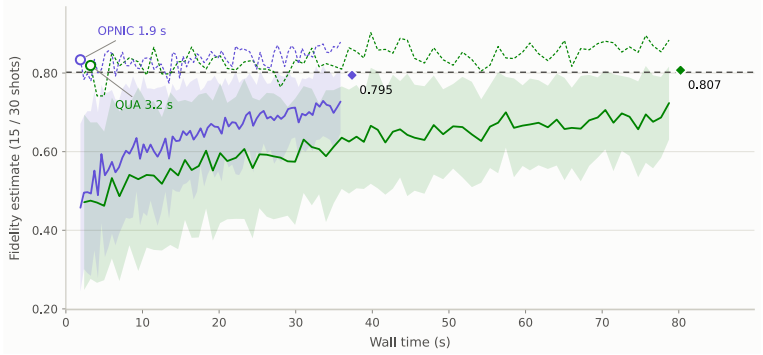
q3–q4

30 / 30 training shots · 1 / 3 seeds



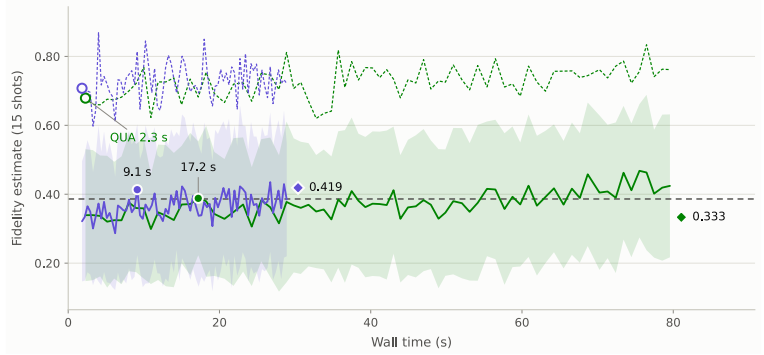
q4–q5

15 / 30 training shots · 3 / 3 seeds



q5–q6

15 / 15 training shots · 3 / 3 seeds



Horizontal axis: wall time (s) · vertical: Bell-state fidelity estimate · axis ranges differ by panel

Fidelity (API / OPNIC)

0.869 / 0.901

Manual reference 0.866

Fidelity (API / OPNIC)

0.807 / 0.795

Manual reference 0.802

Fidelity (API / OPNIC)

0.333 / 0.419

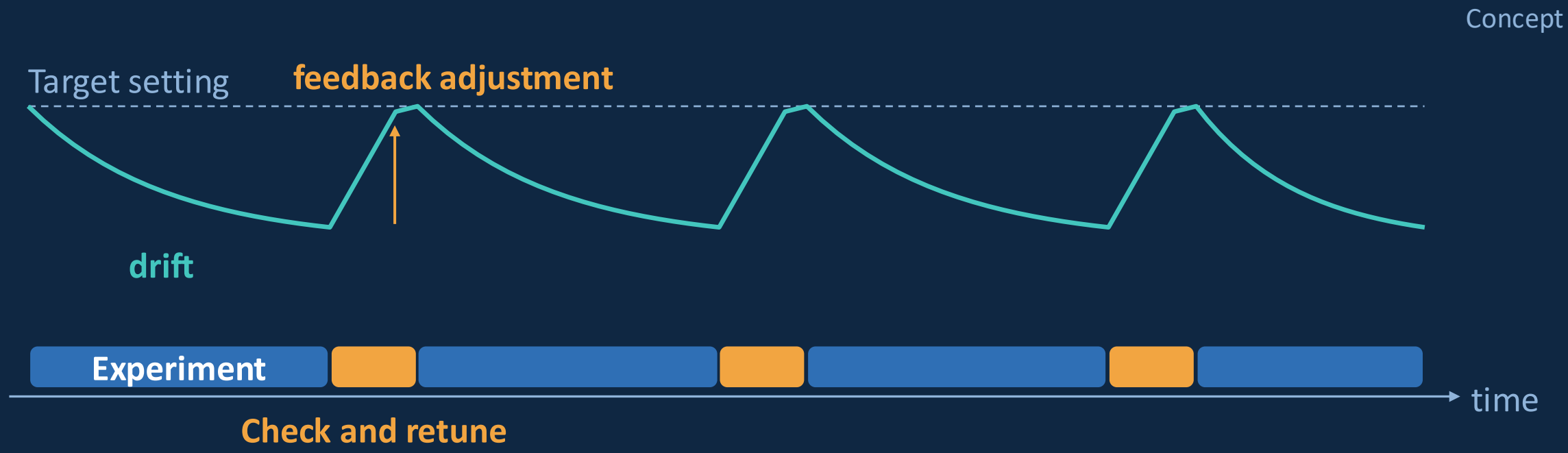
Manual reference 0.386

speed up ≈ 2.1-2.75 x

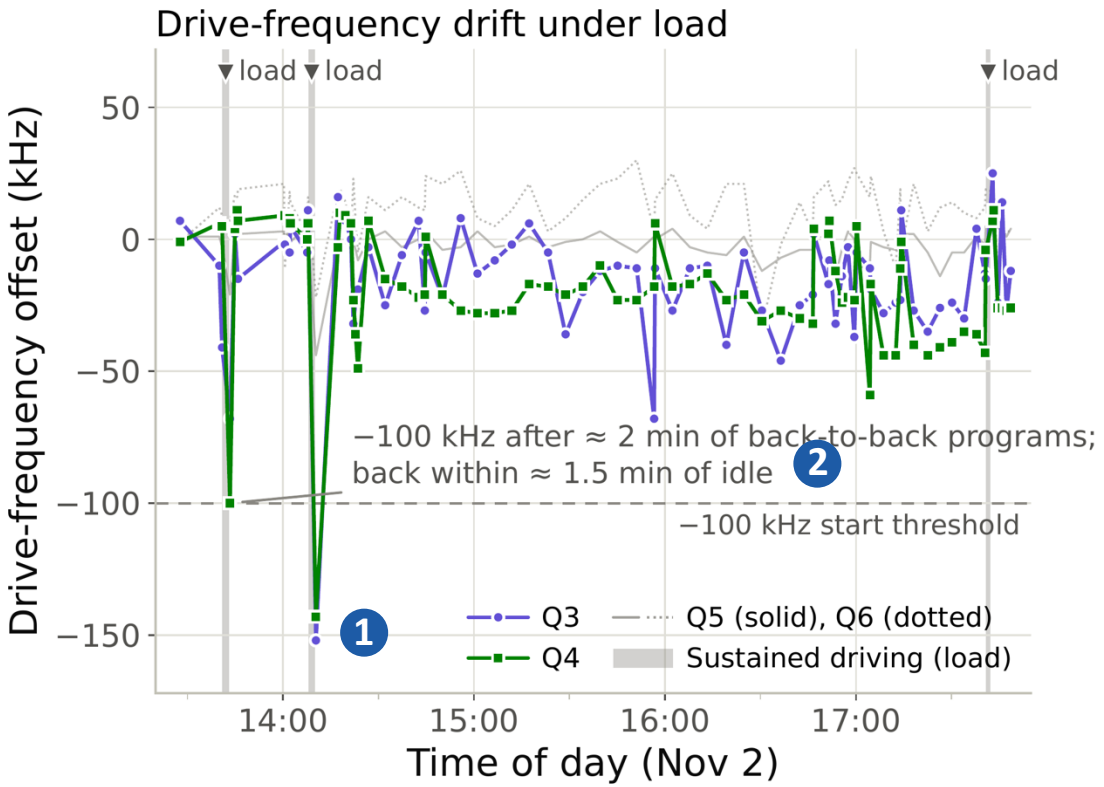
Calibration During Operation

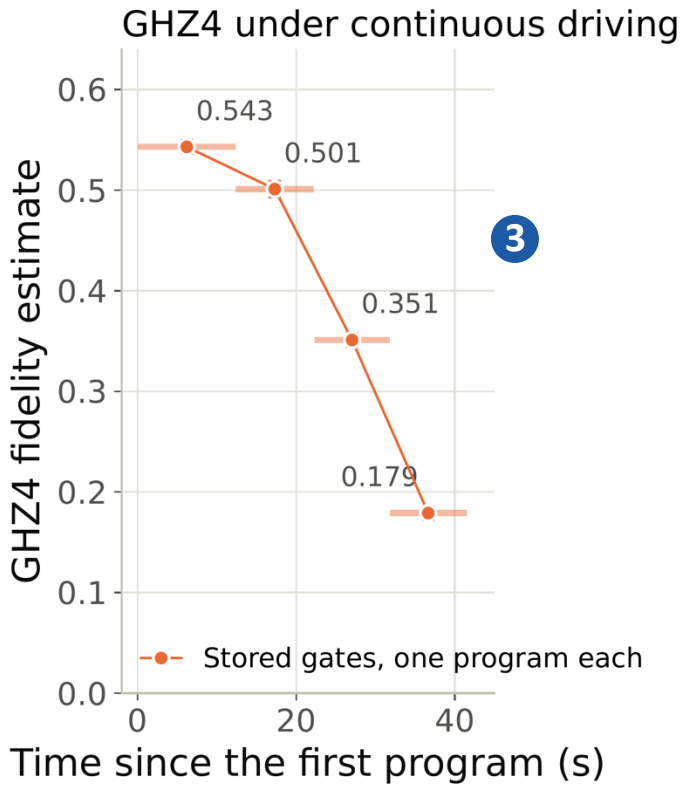
Keeping control settings aligned with a changing device

Frequent checks can reduce the need for manual intervention



Operation Changes the Calibration Point





?

Physical cause unresolved

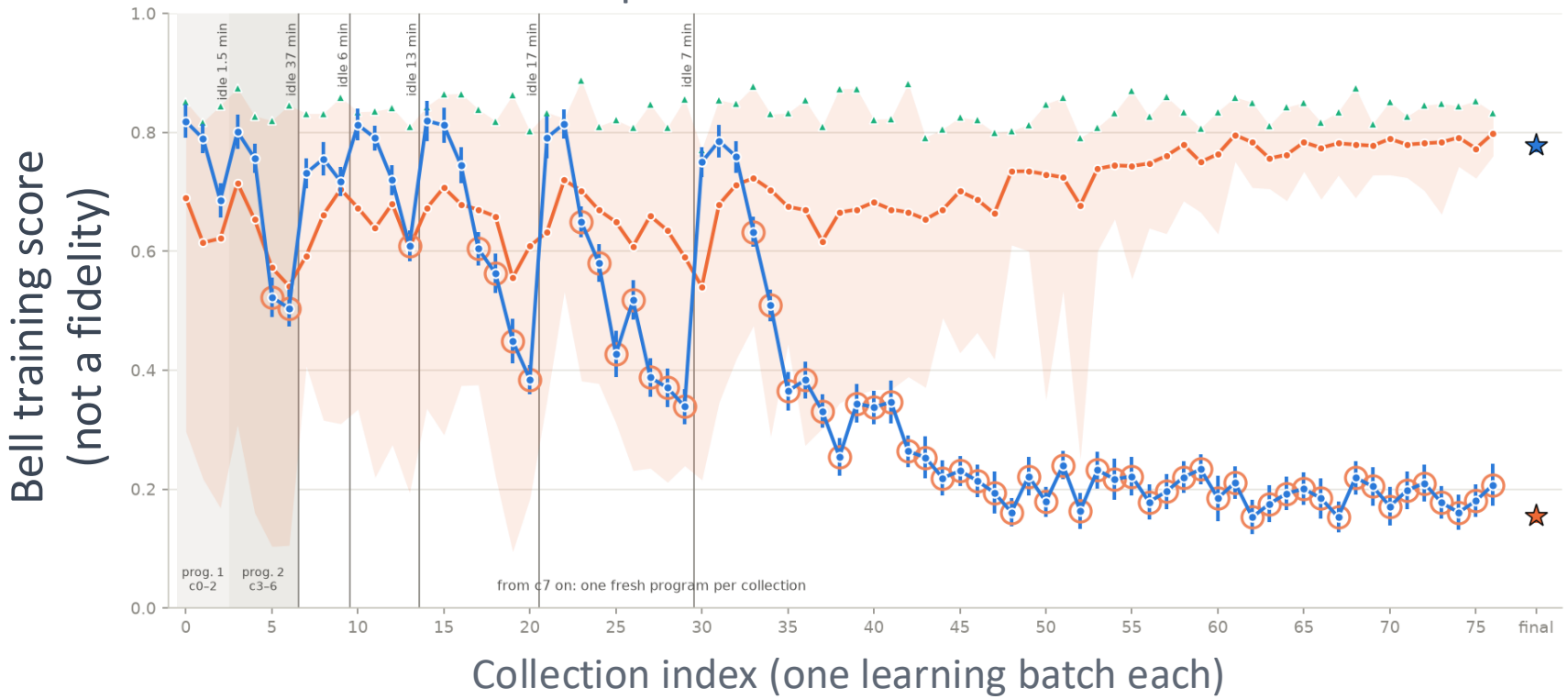
1 Offsets of roughly −70 to −150 kHz under load

2 One observed recovery: about 1.5 min of idle

3 GHZ estimate: 0.54 to 0.18 over 45 s in the shown sequence

The Agent Adapts to the Device Under Load

Historical three-control Bell experiment on Q4–Q5



After rest (★ held-out)

Stored ≈0.78

3 After rest, the advantage reverses

Frozen agent ≈0.16

stored gate (fixed baseline row) ± SE
policy mean (19 samples)
policy samples: min–max of the 19
best sample of the collection
guard-band flag |z| > 3 (recorded)
idle > 60 s

1 Under load: agent score ≈0.8, stored score ≈0.2

2 The search narrows as training progresses

Scaling the Test to Four Qubits

Results and remaining limitations



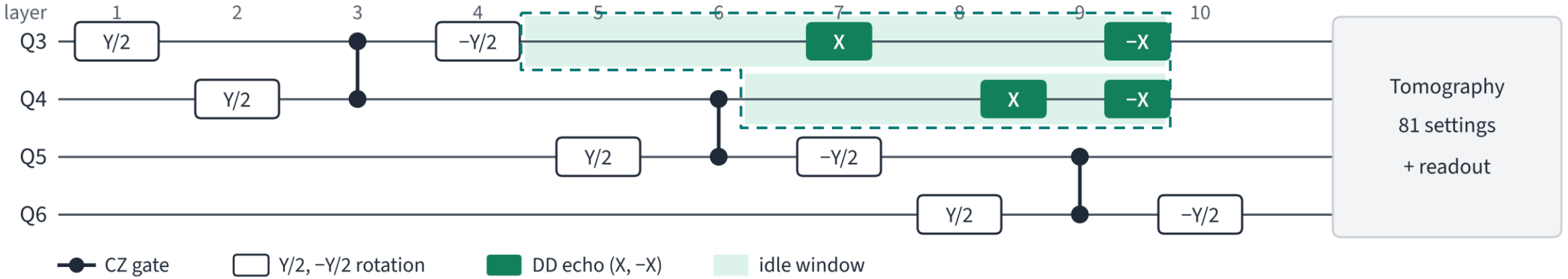
Concept

GHZ target

$(|0000\rangle + |1111\rangle)/\sqrt{2}$

Echo Pulses Help Preserve Four-Qubit Coherence

Echo pulses protect waiting qubits



Procedure

Recalibrate → **Add echoes** → **Optimize seven controls** → **Evaluate**

+4-7 percentage points

GHZ fidelity estimate after adding echoes, across the shown tests

54.9% ± 0.7 percentage points

Single held-out GHZ fidelity estimate; counting SE

GHZ Fidelity Difference After the Rested Run

Frozen agent minus baseline, in the four-qubit GHZ fidelity estimate

+4.3 ± 0.5

percentage points
vs stored settings

+0.3 ± 0.5

percentage points
vs recalibrated settings

Within counting uncertainty

Next Steps Toward Adaptive Calibration

Planned

1 Improve the pulses

Reduce leakage and characterize drift

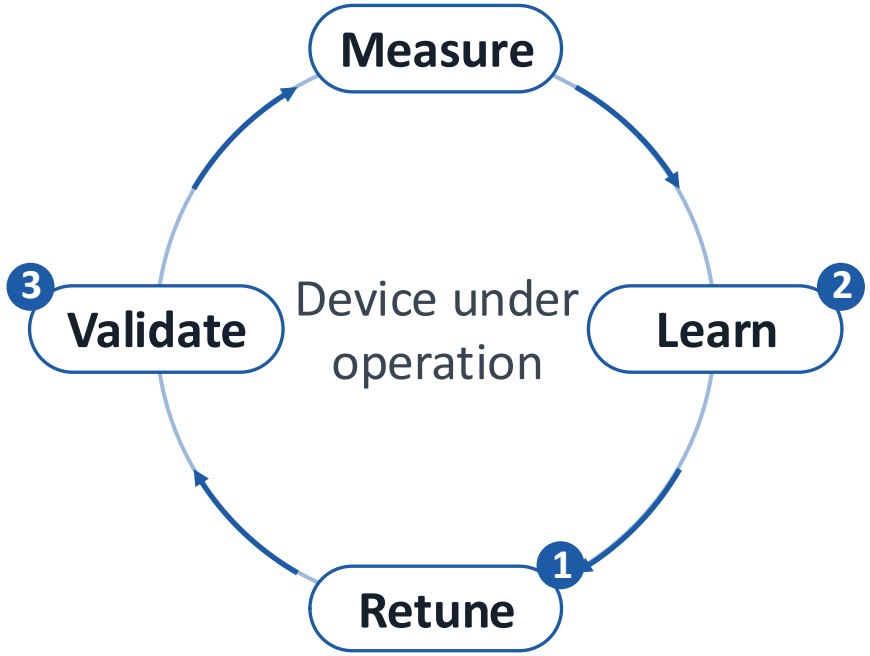
2 Expand the agent

Add useful controls and optimize under operating load

3 Automate the workflow

Coordinate setup, validation, and retuning with an LLM agent

Badges on the loop show where each workstream acts.



Shorter feedback delays on NVQLink with OPNIC + GH200

Take Away

25.9×

faster calibration workflow

Manual → agent

19.4 min → 45 s

≈100–120×

faster communication

NVQLink with OPNIC + GH200

≈2.1-2.7× faster complete runs versus remote API— 83.7 s → 42.1 s, equal-weight group means

Calibrate for the conditions in which the device runs.

Acknowledgments

Academia Sinica and QC-Test

QPU access and technical support

Chii-Dong Chen, Chung-Ting Ke, and David T. Lee

Guidance and support

National Center for High-performance Computing (NCHC)

GPU computing resources

Quantum Machines and Qblox

Technical support

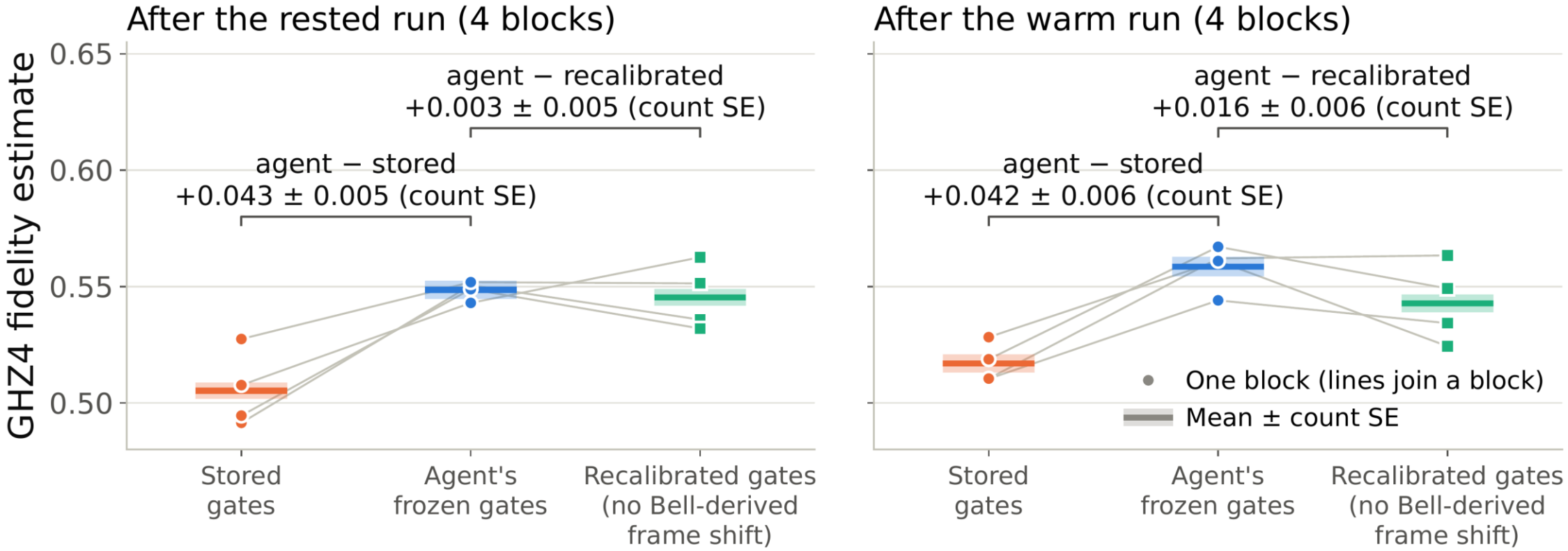
NVIDIA

Ising Calibration and NVQLink technologies

Thank you

Appendix

The interleaved test: the agent's gates beat the stored gates; recalibrated gates did as well



Four blocks; the agent was measured in the middle of every block; order effects unresolved.

- **Rested run:** the agent's frozen gates scored +0.043 ± 0.005 over the stored gates (4 of 4 blocks).
- **Against recalibrated gates** without the Bell-derived frame shift: +0.003 ± 0.005, no difference detected.
- **Warm run:** +0.042 ± 0.006 over the stored gates, and +0.016 ± 0.006 over the recalibrated gates.
- **Reading:** the agent recovered, inside the GHZ circuit, a loss that a Bell-derived frame shift had caused.

± is the count SE; four blocks; the agent was measured in the middle of every block, and the order effects are unresolved.

Proctor et al., Nat. Commun. 11, 5396 (2020).

Manual calibration of one Bell pair: 12 node types, ≈ 43 min per clean pass (estimate)

schematics

1 Resonator spectroscopy
Sets the readout frequency.



≈ 3 min (run ≈ 1 estimated + analysis ≈ 2)

2 Qubit spectroscopy
Finds the qubit transition.



≈ 4 min (run ≈ 2 estimated + analysis ≈ 2)

3 Flux sweet spot
Sets the idle flux bias, where the qubit is first-order insensitive to flux noise.



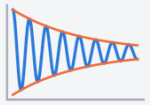
≈ 5 min (run ≈ 2 estimated + analysis ≈ 3)

4 Rabi
Sets the π-pulse amplitude.



≈ 1.5 min (run ≈ 0.5 estimated + analysis ≈ 1)

5 Ramsey
Fine-tunes the drive frequency and gives T2*.



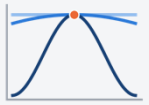
≈ 1.3 min (run 0.2–0.3 measured + analysis ≈ 1)

6 Readout optimization
Sets the readout point and the discrimination threshold.



≈ 2.5 min (run 0.1 measured + ≈ 1.5 estimated for frequency and power, + analysis ≈ 1)

7 Amplitude fine-tuning
Sets the π and π/2 amplitudes by error amplification.



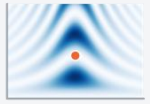
≈ 6 min (run 3.2 + 1.6 measured, for four qubits + analysis ≈ 1)

8 Coupler idle point
Sets the coupler where the residual ZZ interaction ≈ 0.



≈ 4 min (run ≈ 2 measured, a coupler-flux scan + analysis ≈ 2)

9 CZ chevron
Sets the flux-pulse amplitude and duration of the |11⟩ exchange with the doubly-excited level.



≈ 5.5 min (run ≈ 2.5 estimated + analysis ≈ 3)

10 Conditional phase
Fine-tunes the qubit and coupler amplitudes for a conditional phase of π with low leakage.



≈ 5.5 min (run 1.5–3 measured + analysis ≈ 3)

11 Single-qubit phase corrections
Sets the two virtual-Z frame updates after the CZ.



≈ 1.5 min (run 0.2–1 measured + analysis ≈ 1)

12 Checks
Readout confusion matrix, Bell-state tomography and randomized benchmarking.



≈ 3 min (run 0.2–1 measured + analysis ≈ 2)

One clean pass **≈ 43 min** (estimate: ≈ 20 min of run time + ≈ 22 min of analysis).

With the usual re-runs and amplitude ↔ phase iterations **≈ 1–1.5 h** (estimate).

Measured re-calibrations on our device: **6.3 min** scripted (cards 5, 6, 7, 11), and **7.6 min** with each fit read before the next (cards 5–7).

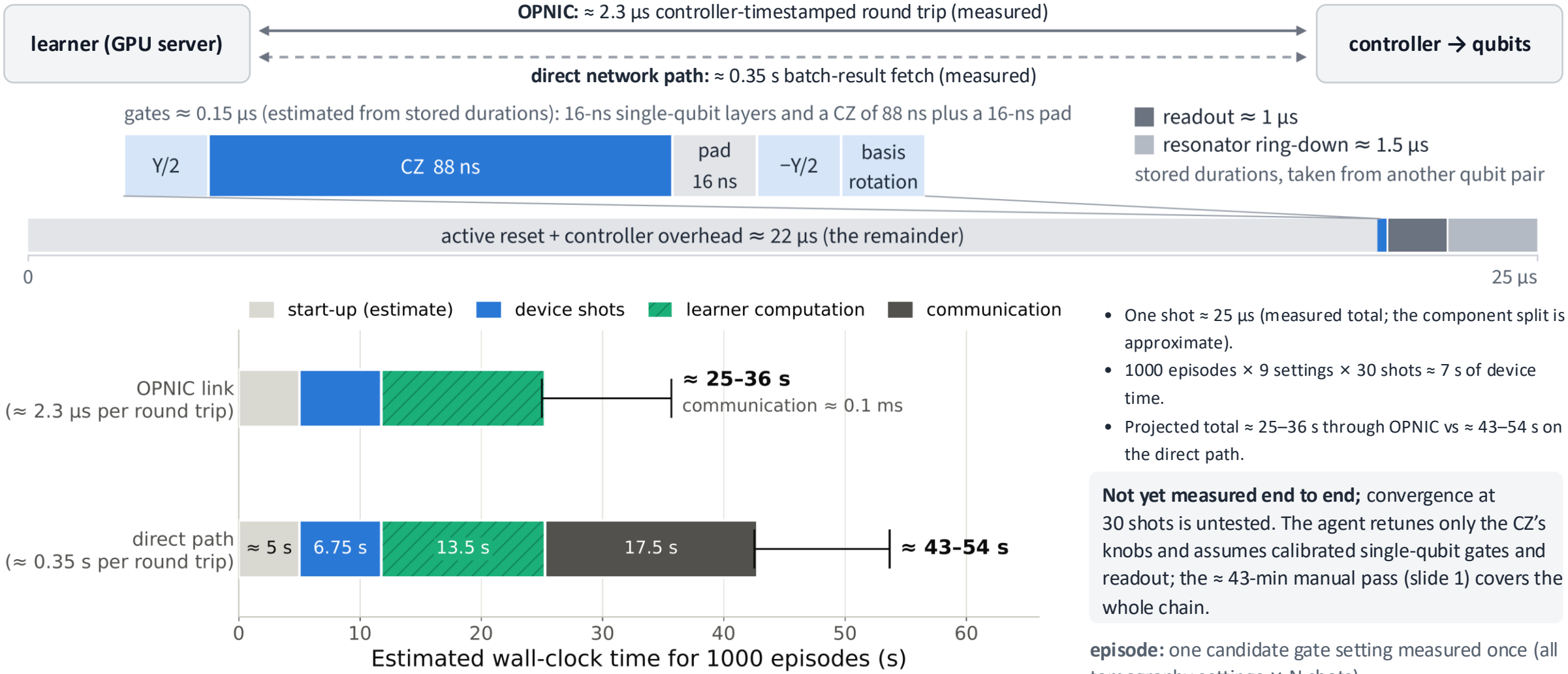
Node: one calibration routine, a scan plus a fit.

Run: the node's wall time from launch to saved result (connect, compile, execute, fit, save). It is measured where marked; the other run times are estimates from each node's default scan.

Analysis: an expert reading the fit and deciding whether to accept, change the range or re-run. All analysis times are estimates.

Kelly et al., arXiv:1803.03226 (2018); Krantz et al., Appl. Phys. Rev. 6, 021318 (2019).

Projected: retuning the CZ knobs with 1000 episodes through OPNIC would take ≈ 25–36 s



Built from measured unit costs. Hardware Bell runs (300 shots per setting) first reached a training reward of 0.85 within ≈ 230–680 episodes; a 30-shot simulation reached 0.90 within ≈ 1000–1150 episodes.

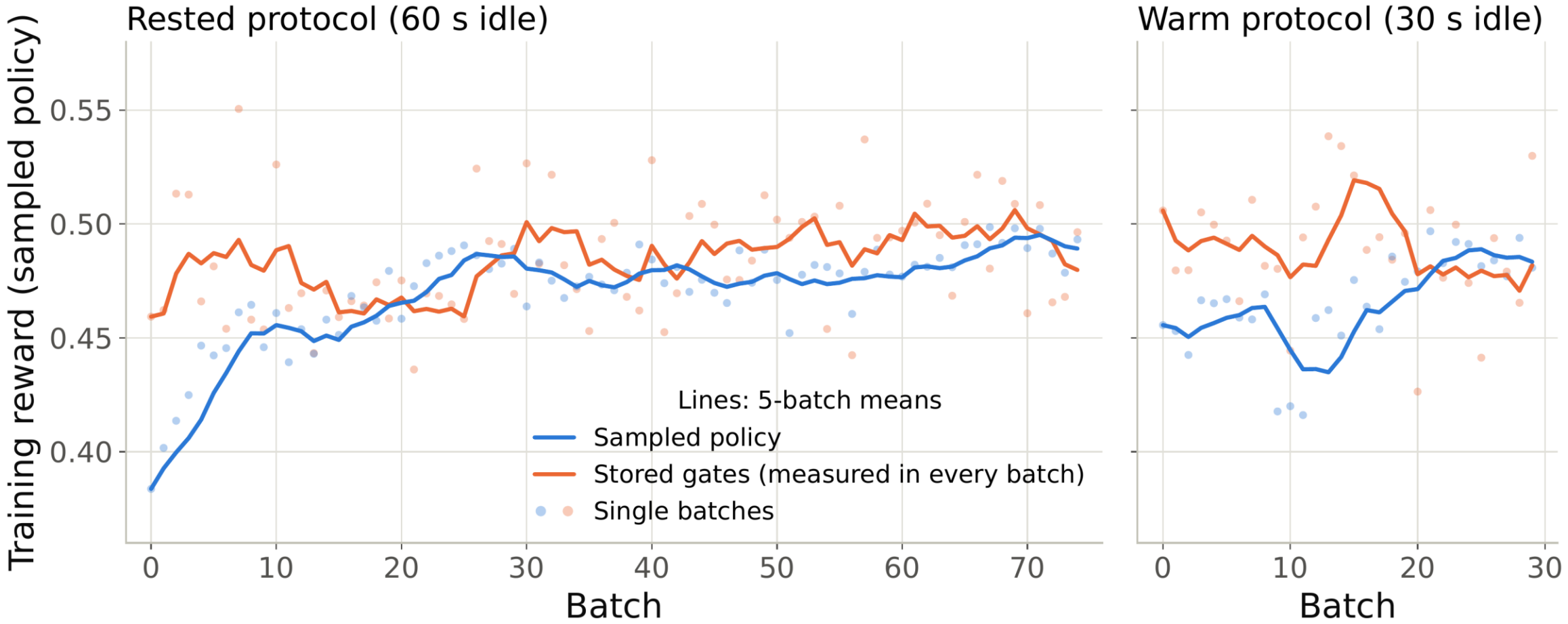
- One shot ≈ 25 µs (measured total; the component split is approximate).
- 1000 episodes × 9 settings × 30 shots ≈ 7 s of device time.
- Projected total ≈ 25–36 s through OPNIC vs ≈ 43–54 s on the direct path.

Not yet measured end to end; convergence at 30 shots is untested. The agent retunes only the CZ's knobs and assumes calibrated single-qubit gates and readout; the ≈ 43-min manual pass (slide 1) covers the whole chain.

episode: one candidate gate setting measured once (all tomography settings × N shots)
OPNIC: a low-latency link between the controller and the GPU server

Ristè et al., Phys. Rev. Lett. 109, 240502 (2012); Reuer et al., Nat. Commun. 14, 7138 (2023).

Two protocols for the agent: warm (under the working load) or rested (the idle device)

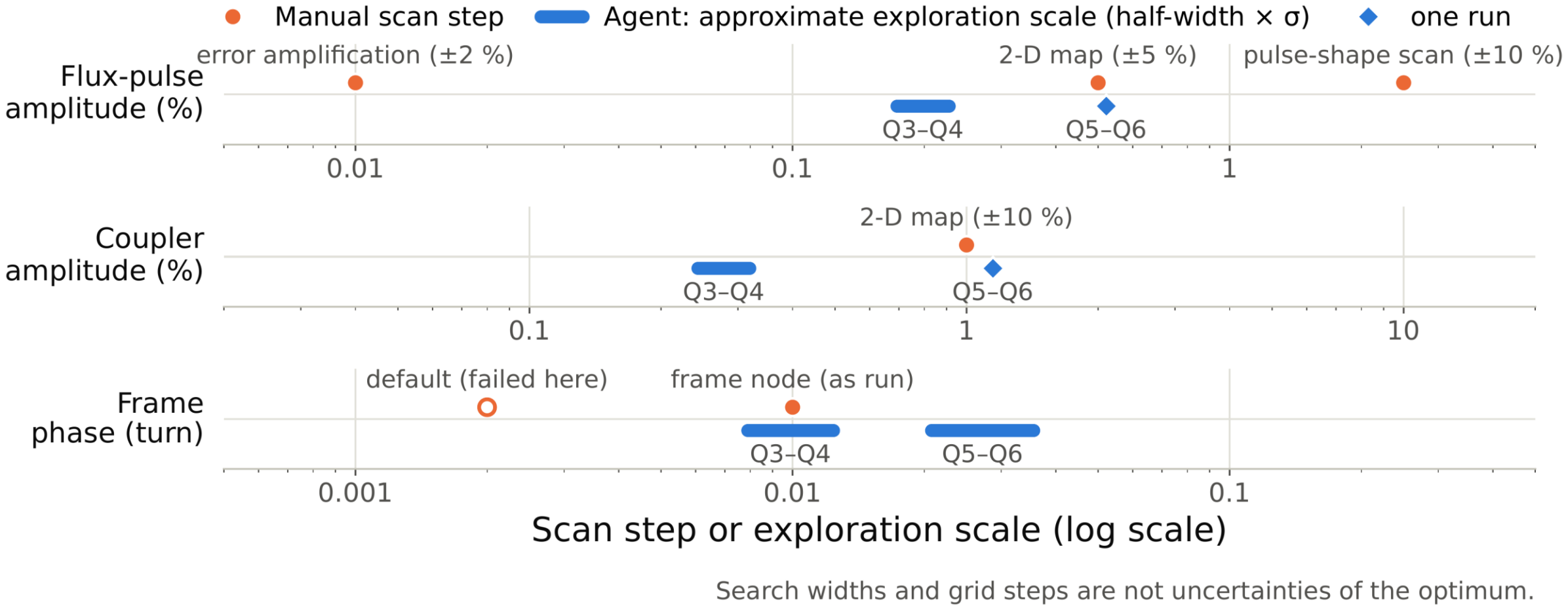


- **Rested** (60 s idle between batches; 75 batches; GHZ state, 7 knobs): optimizes for the idle device.
- **Warm** (30 s idle; 30 batches): dynamical calibration under the working duty.
- **Both climbed** to the stored gates' level within ≈ 20 batches.

- **Both moved the same knob the same way:** the Q5–Q6 control frame, +0.021 turn (rested) and +0.032 turn (warm).
- **Choose by goal:** warm for the working duty, rested for the idle optimum.

Proctor et al., Nat. Commun. 11, 5396 (2020); Kelly et al., arXiv:1803.03226 (2018).

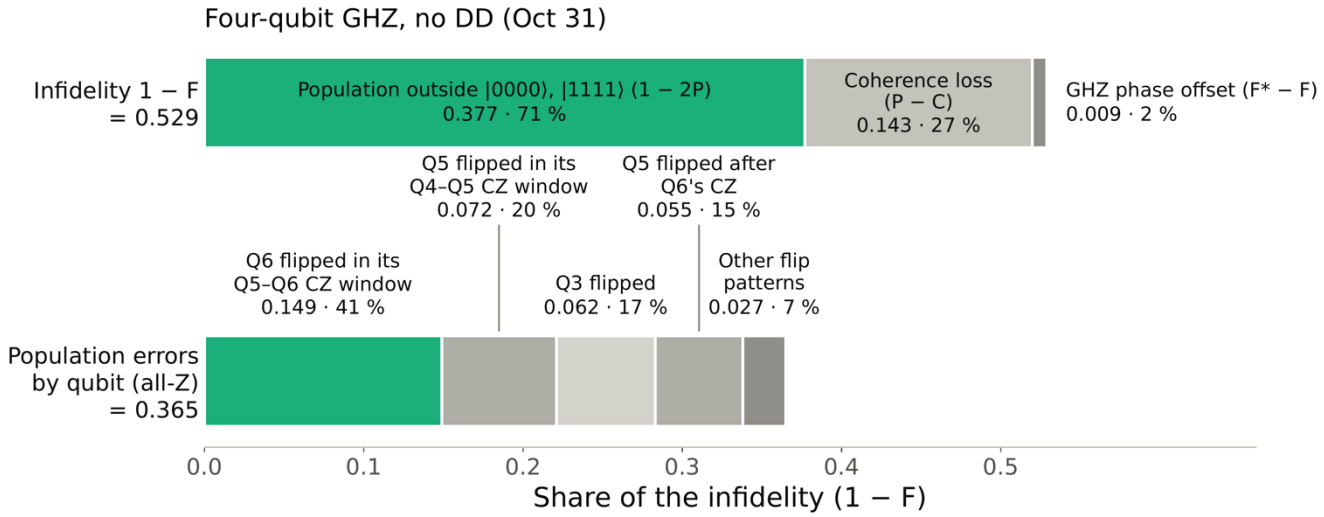
Resolution: manual scan steps vs the agent's final exploration scale



- **Manual steps:** error-amplification nodes step 0.01 % in amplitude, and the frame node ran at 0.01 turn.
- **The agent ends at an exploration scale of** 0.17–0.52 % (flux), 0.24–1.15 % (coupler) and 0.008–0.036 turn (phase).
- **The agent's edge is not finer steps:** it tunes the circuit's own score, in place.

Where the four-qubit GHZ infidelity goes, and what the agent can fix

One measured day, without DD: 1 − F ≈ 0.53 = 71 % population errors + 27 % coherence loss + 2 % phase offset; a split by component, not by physical mechanism.



Error source	Measured	Fixable with the agent's knobs?
Q6 flips inside the Q5–Q6 CZ window	0.149 of 0.365 population errors	Partly: its frame phase is a knob
Leakage out of |11⟩ (Q5–Q6 CZ)	≈ 10–20 % of |11⟩ per gate (inferred)	Partly: new pulse shapes are candidates
Idle dephasing of Q3, Q4	DD recovered +0.04 to +0.07	No: needs DD in the circuit
Frame-phase errors	≈ 0.05-turn offsets; drift 0.03–0.15 turn per hour	Yes: the agent re-locks them (+0.04)
Q6's coherence window	contrast 0.56–0.59 (ideal 1); mechanism unresolved	No: not addressed by the present knobs
Load-induced drift	−0.36 within 45 s of driving	No: rest, or re-calibrate fast

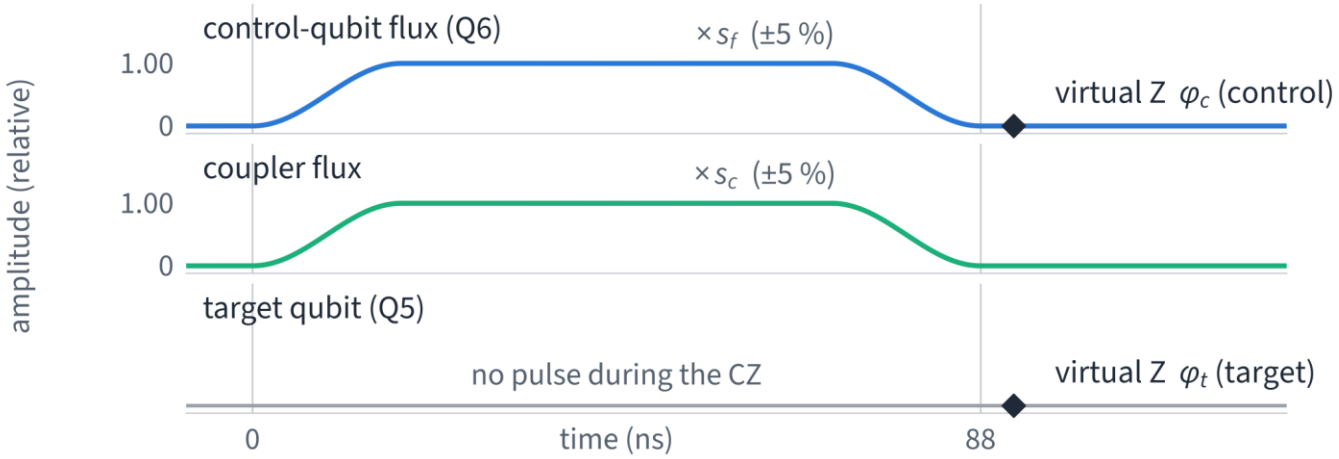
Different experiments; overlapping effects; not an additive error budget by mechanism.

DD: dynamical decoupling, echo pulses on idle qubits · **GHZ state:** the four-qubit entangled state (|0000⟩ + |1111⟩)/√2

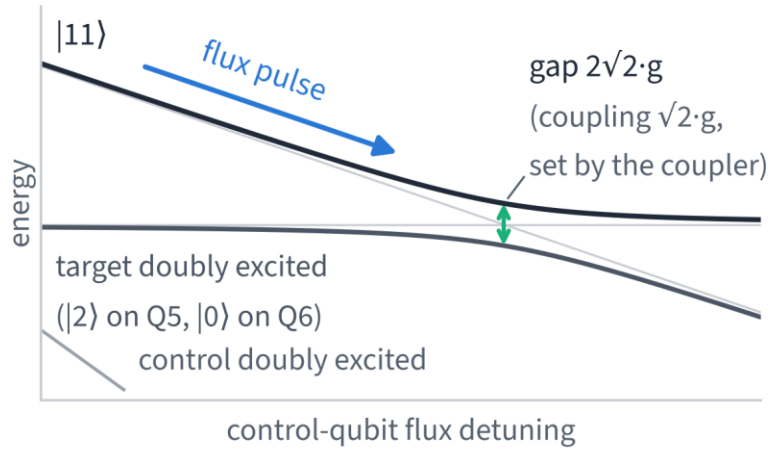
Wood & Gambetta, Phys. Rev. A 97, 032306 (2018); Müller et al., Rep. Prog. Phys. 82, 124501 (2019); Krinner et al., Phys. Rev. Applied 14, 024042 (2020).

The CZ gate: one flux pulse, one coupler pulse, two frame corrections

CZ pulse sequence



Two-excitation levels



schematic: pulse edges and energy levels illustrative

How it works

- **Flux pulse:** tunes the control qubit until |11⟩ meets a doubly-excited level.
- **Coupler pulse:** sets the coupling √2·g; one full exchange gives the CZ phase π.
- **Frame corrections:** virtual-Z updates ϕ_c and ϕ_t cancel each qubit's phase; they take no time.

Manual calibration, ≈ 20 min per clean pass (estimate)

each step as run + analysis

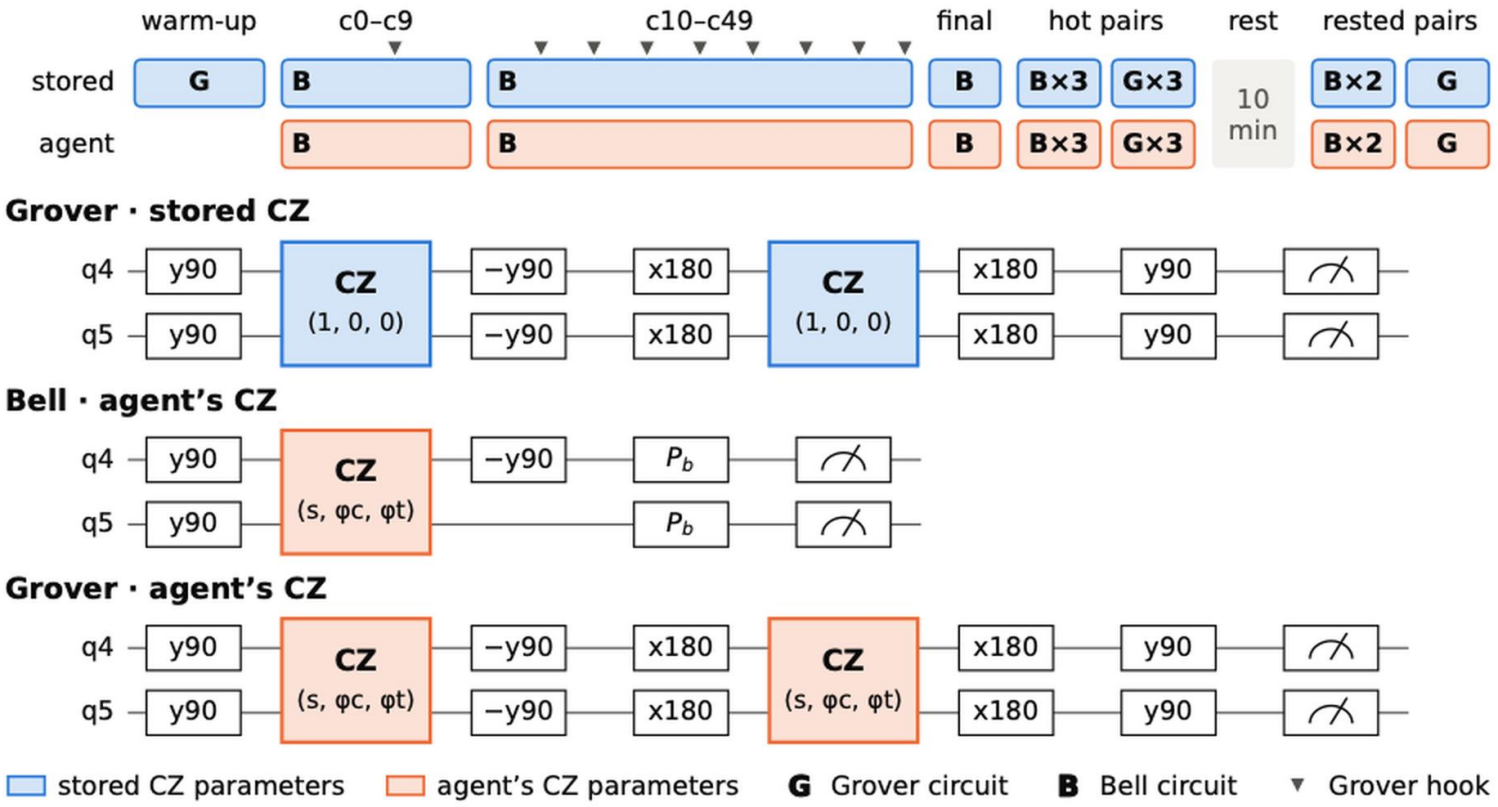
1 Coupler idle point (ZZ ≈ 0)	≈ 4 min (≈ 2 measured + ≈ 2)
2 Chevron, flux amplitude × duration	≈ 5.5 min (≈ 2.5 estimated + ≈ 3)
3 Amplitude–coupler map (phase π, minimum leakage)	≈ 3 min (0.9–1.6 measured + ≈ 1.5)
4 Error-amplified fine-tune of the conditional phase	≈ 2.5 min (0.5–1.2 measured + ≈ 1.5)
5 Frame phases ϕ_c, ϕ_t	≈ 1.5 min (0.2–1 measured + ≈ 1)
6 Check (readout confusion + Bell tomography)	≈ 3 min (0.2–1 measured + ≈ 2)

Total: ≈ 20 min per clean pass, ≈ 40 min with the usual iterations (estimates; all analysis times estimated). Step 4 resets the frame phases, so step 5 must follow it every time.

DiCarlo et al., Nature 460, 240 (2009); Yan et al., Phys. Rev. Applied 10, 054062 (2018); Sung et al., Phys. Rev. X 11, 021058 (2021); Koch et al., Phys. Rev. A 76, 042319 (2007); McKay et al., Phys. Rev. A 96, 022330 (2017).

Task 2. Grover: Agent-Calibrated vs Stored Parameters

Protocol and circuits: train on Bell, test the same CZ settings on a two-CZ Grover circuit



TRAINING

The agent learns on the **Bell** circuit only (rows B). **Grover** (G) is a held-out circuit that uses the CZ twice.

SIDE BY SIDE

Every phase after the warm-up runs both gate sets: **stored** CZ = (1, 0, 0) and the **agent's** CZ = (*s*, *φc*, *φt*).

SEQUENCE

Warm-up → training c0–c49 (▼ Grover hooks) → final Bell → hot pairs → 10-min rest → rested pairs.

SCORE

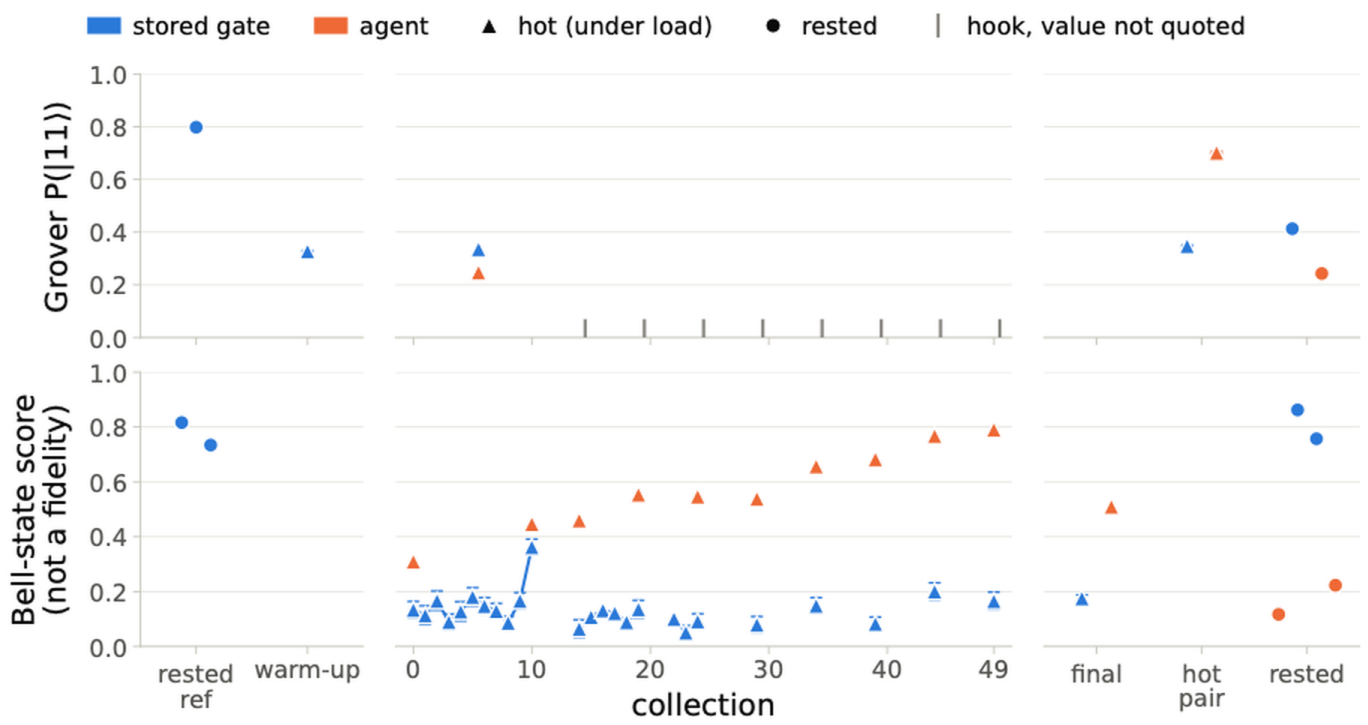
The ideal Grover output is |11⟩ with probability 1, so P(|11⟩) is the success measure.

TAKEAWAY *Grover asks whether the Bell-trained CZ is a better gate, not just a better Bell score.*

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Task 2. Grover: Agent-Calibrated vs Stored Parameters

Results: Grover P(|11⟩) (top) and Bell score (bottom) for both gate sets · hardware measurement



▲ hot (under load) · ● rested · values read from the figure (≈)

Grover P(|11⟩), hot pair

CZ	raw	RO corrected
Stored (1, 0, 0)	0.301 ± 0.003	0.348 ± 0.003
Agent (frozen)	0.590 ± 0.003	0.702 ± 0.004
Agent – stored	+0.289 ± 0.004	+0.354 ± 0.005

Agent's frozen CZ: (*s*, *φc*, *φt*) = (0.956, −0.059, −0.051), φ in turns · 30,000 shots per row

UNDER LOAD (HOT PAIR)

The agent's CZ doubles the Grover success — on a circuit it never trained on.

AFTER A 10-MIN REST

Stored ≈0.42 vs agent ≈0.25 on Grover; Bell flips the same way.

TAKEAWAY *Under load the Bell-trained CZ also wins on Grover; after a 10-minute rest the stored CZ wins again.*